

Engineering Physics (2025)

Course code 25PY101

Module 2 Unit 3: Lasers

Course Instructor:

Dr. Sreekar Guddeti

Assistant Professor in Physics

Department of Science and Humanities

Vignan's Foundation for Science, Technology and Research

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Laser

Optical fibre

Laser

Optical fibre

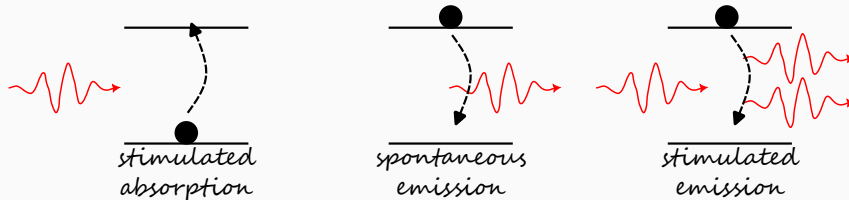
Laser

Laser

Optical fibre

Laser – Three Quantum Processes

- L.A.S.E.R is acronym for Light Amplification by Stimulated Emission of Radiation.
- Einstein predicted the possibility of lasing in 1916 by introducing concept of stimulated emission.



- Light interacts with matter by three quantum processes
 - Stimulated absorption
 - Spontaneous emission
 - Stimulated emission

Laser – Three Quantum Processes (contd...)

Definition

Stimulated absorption occurs when electron transitions from lower energy level to higher energy level upon the absorption of photon.



Definition

Spontaneous emission occurs when electron transitions from higher energy level to lower energy level by the emission of photon.



Definition

Stimulated emission occurs when electron transitions from higher energy level to lower energy level in the presence of a photon by the emission of additional photon.

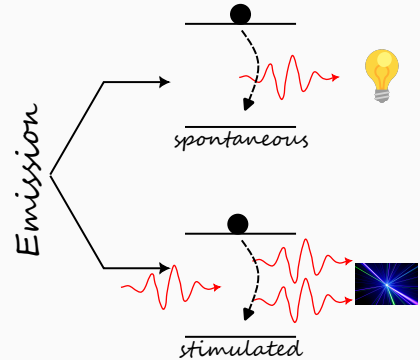


Lasing action

Definition

Lasing action is defined as the condition when the stimulated emission dominates the spontaneous emission process.

- Emission of radiation when the electron makes a transition from higher energy level to lower energy level can be either
 - spontaneous
 - stimulated
- In normal light 💡, the spontaneous emission dominates stimulated emission.
- In laser light 🌈, the stimulated emission dominates spontaneous emission.



Laser – Conditions and requirements for lasing action

From the Einstein's theory ¹ of light-matter interaction, the conditions for lasing action are –

- high photon density,
- large lifetime of excited levels, and
- high population of excited levels to achieve lasing action.

To satisfy these three conditions for lasing action, the requirement of lasing action are –

1. Population inversion
2. Meta-stable energy level
3. Radiation confinement

¹Refer to Chapter 44 Avadhanulu for a discussion on this theory.

Thermal equilibrium and normal state

Definition

Maxwell-Boltzmann statistics says the probability of occupying higher energy level is lower than the probability of occupying lower energy level. The probability of occupying energy level E is proportional to

$$P_{MB} \propto \exp\left(-\frac{E}{k_B T}\right)$$

Definition

Normal state is defined as the state under thermal equilibrium.

- Under the conditions of thermal equilibrium, the occupancy of energy levels is governed by Maxwell Boltzmann statistics.



Problem

Find the ratio of populations of two states in a He–Ne laser that produces light of wavelength 6328 Å at 27 °C.



normal state

1. Population inversion and inverted state

Definition

Population inversion is defined as the condition when the number of electrons in the higher energy level is more than the number of electrons in the lower energy level.

Definition

Inverted state is defined as the state attained after population inversion.

- Since, under thermal equilibrium, the number of electrons in higher energy level cannot be more than the number of electrons in lower energy level, the inverted state is a **non-equilibrium state**.



Problem

A 10 mW *He-Ne laser* has efficiency of 1 %. Assume that all input energy is utilized in pumping the atoms from ground state to the excited state, which is 20 eV above the ground state. Find how many atoms are promoted to the excited state per second.



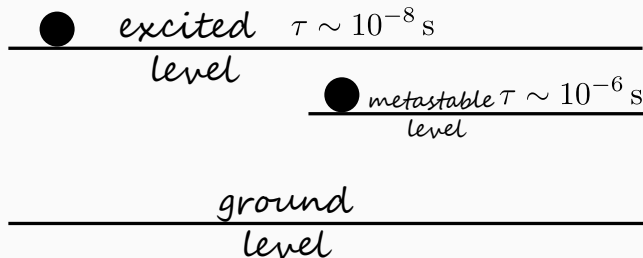
inverted state

2. Meta-stable energy level

Definition

Meta-stable energy level is an excited energy level with an unusually long life time.

- Excited energy levels have a very low lifetime of the order of 10 ns.
- However, some excited energy levels can have larger lifetimes due to various reasons. These are called meta-stable energy levels.
- These states lie in between excited energy levels and ground energy level both in terms of lifetime and energy.
- The meta-stable energy level lifetime is of the order of 10^{-3} s to 10^{-6} s.

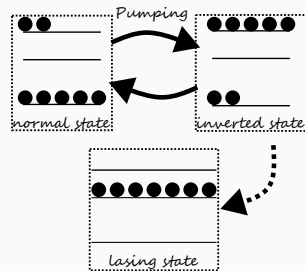


Meta-stable energy level $\implies$ lasing state

Definition

Lasing state is defined as the condition when the number of electrons in the meta-stable level is more than the number of electrons in the ground level.

- If the lifetime of electron in meta-stable level is 10^{-3} s to 10^{-6} s, then the electron stays nearly 10^2 to 10^5 times longer in the meta-stable level than in the excited level.
- **Analogy:** To understand the scale difference of lifetimes, if we consider lifetime of excited level as 1 s, then lifetime of meta-stable level will be as 100 s to 100 000 s i.e. 2 minute to 1 day.
- If the pumping is continued, due to difference in time scale, spontaneous emission from the excited level significantly populates the meta-stable level. This state is called the **lasing state**.



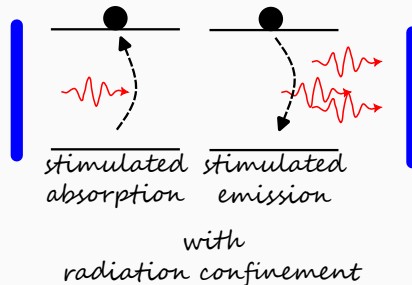
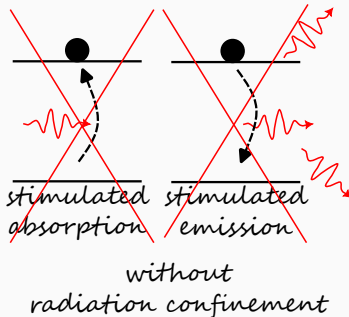
3. Radiation confinement

Definition

Radiation confinement is the one of the conditions to achieve lasing action, where the photons are confined to a region in space and time.

The advantage of radiation confinement is two-fold –

- If the radiation is confined within a region, then the electrons can undergo stimulated absorption and lead to higher population of excited level.
- The presence of photons can lead to stimulated emission leading to lasing action.



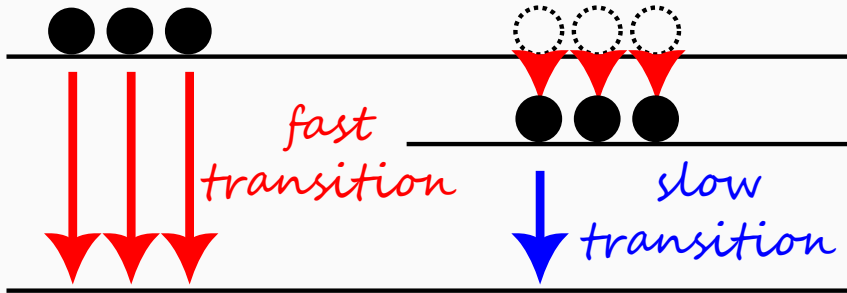
To satisfy the above three requirements of lasing action, we need three components to any laser –

1. Active medium
2. Optical pump
3. Optical resonant cavity

1. Active medium

Definition

Active medium is the physical medium where lasing action takes place. The medium hosts the meta-stable level that has larger lifetime so that population inversion can be sustained.

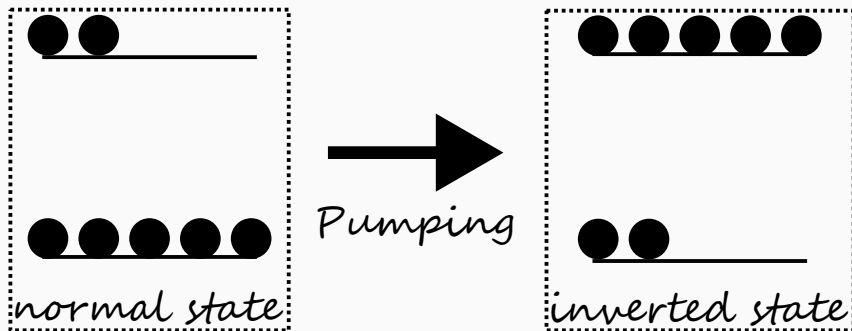


2. Optical pump

Definition

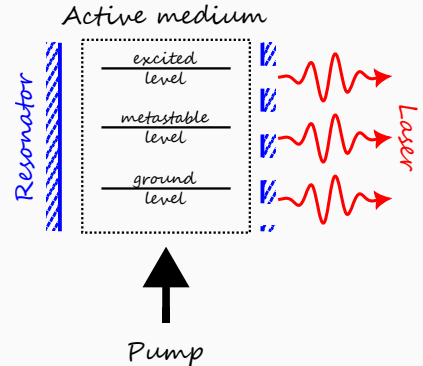
Optical pump is the external agent that transforms the active medium from normal state to inverted state.

- Optical pump provides the energy to raise the electrons from ground state to the excited so that population inversion takes place.



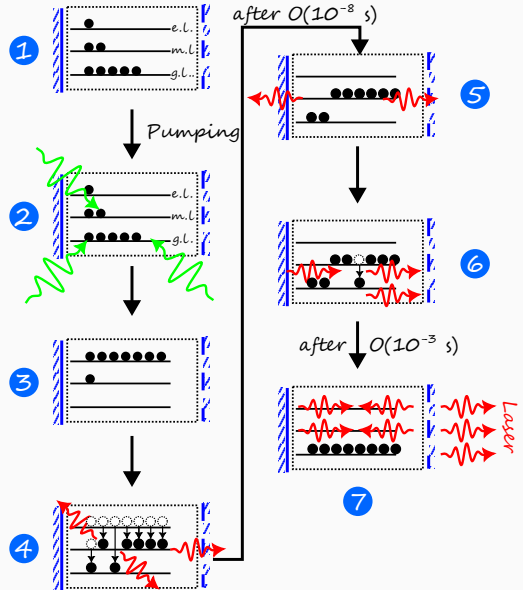
3. Optical resonator

- In order to confine the radiation within the active medium, the reflection property of light is used.
- A pair of optically flat/plane mirrors enclose the active medium so that the photons get confined.
- In order to extract the photons from a single direction, one mirror is made fully reflecting whereas second mirror is partially reflecting. This arrangement is called optical resonant cavity or simply optical resonator.
- The commonly used optical resonator is called Fabry-Perot interferometer.



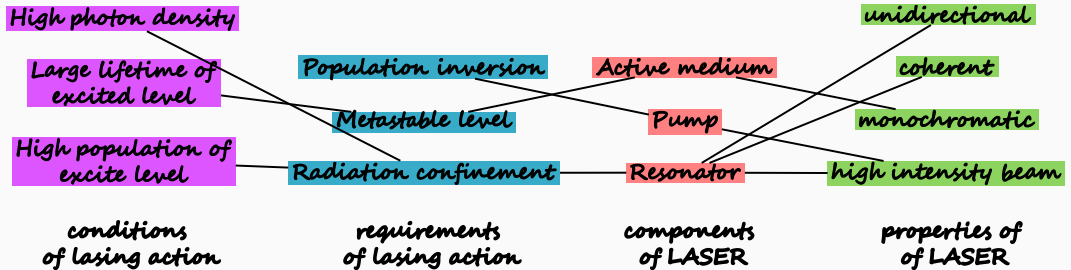
Lasing action

1. Normal state
2. Stimulated absorption
3. Inverted state
4. Spontaneous emission
 - most photons along the resonator axis are lost.
 - some photons along the resonator axis are trapped.
 - photons of other color are lost or reabsorbed.
5. Lasing state
6. Stimulated emission
 - Emitted photons further stimulate other transitions leading to **gain/amplification**.
7. L.A.S.E.R.



Lasing action (contd...)

- Lasing action occurs because
 - multiplication of photons leads to high intense beam.
 - a fraction of photons escape the resonator due to partially reflected mirror.
- Lasing action leads to



Laser – characteristics

Laser light 	Ordinary light 
Unidirectional	Non-directional
Coherent	Incoherent
Monochromatic	Non-monochromatic
High intensity	Low intensity
Low pulse width	High pulse width

- Coherent means photons in the beam have same phase
- Monochromatic beam consists of photons of single color/wavelength.

Diode laser – Construction

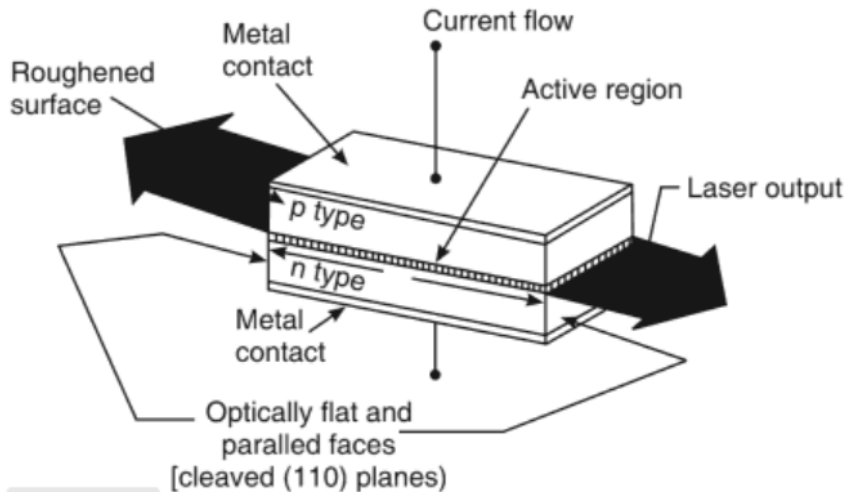
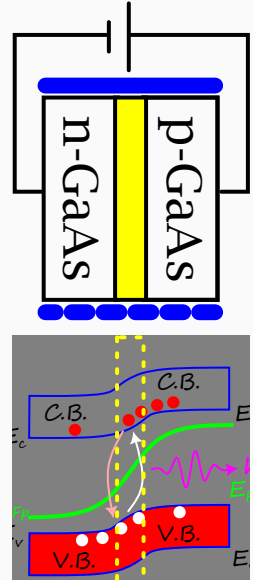


Fig. 44.24:

Schematic of homojunction diode laser

Diode laser – working principle

- In a diode laser
 - Depletion region is the active medium,
 - Forward biasing is the pump, and
 - Polished faces of the GaAs crystal are the optical resonator.
- Since the forward biasing populates the electrons in the n region and holes in the p region, when they meet at the depletion region, the active medium is already in the lasing state.
- Therefore, there is no need for a meta-stable state.



Diode laser – applications

- Industrial precision cutting
- Lasik surgery
- LiDAR (Light Imaging, Detection and Ranging) technology for distance measurement
- Optical fibre communication technology

Optical fibre

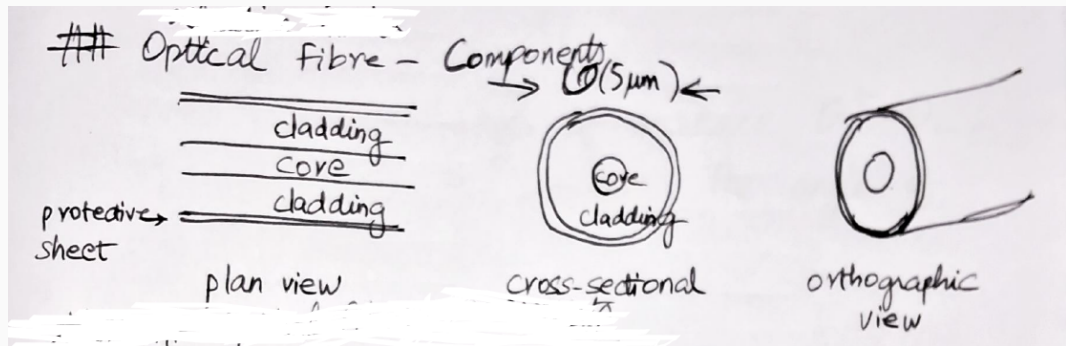
Laser

Optical fibre

Optical fiber – light pipe

- Optical fiber is an opto-electronic device that is the backbone of modern communication systems.
- Useful for long distance communication.
- Harnesses the speed of light and property of light propagation across media with different refractive indices.

Optical fiber – construction



Optical fiber – working principle

- The speed of light varies in different media.
- It is highest in vacuum and is given by $v = c$.
- In other medium, it is quantified by the refractive index n , so that the velocity of light in the medium is

$$v = \frac{c}{n}.$$

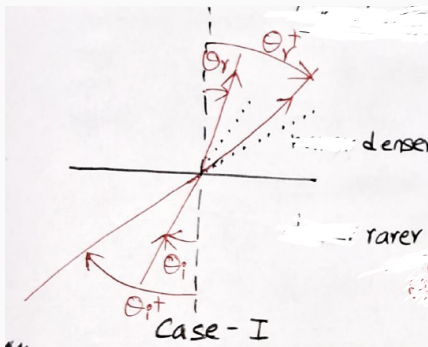
- At the interface of media, speed of light changes leading to two cases.
 - Case I: Light propagates from rarer to denser medium.
 - Case II: Light propagates from denser to rarer medium.

medium	n
air	1
water	1.33
glass	1.5

Table 1: Refractive indices of commonly available media.

Case I: Rarer to denser medium

Consider light is incident at an angle of incidence θ_i with respect to normal.

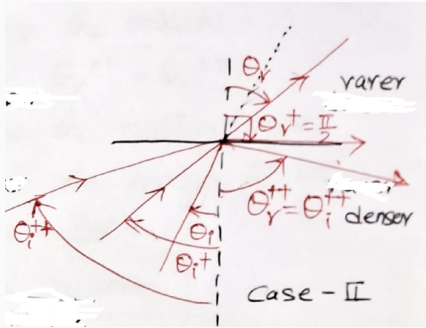


- Since light is faster in rarer medium, and slower in denser medium, it bends towards the normal. Therefore, it refracts.
- Angle of refraction is less than angle of incidence,

$$\theta_r < \theta_i.$$

- Similarly, for larger angle of incidence θ_i^+ , light refracts with angle of refraction $\theta_r^+ < \theta_i^+$.

Case II: Denser to rarer medium



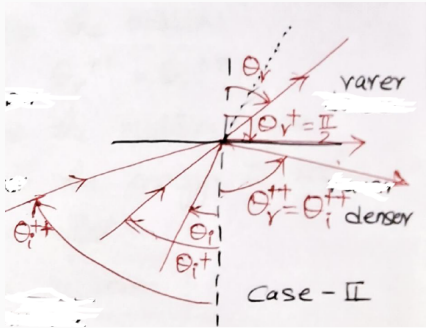
- Since light is slower in denser medium and faster in rarer medium, it bends away from the normal.
- Case (i) Angle of refraction is more than angle of incidence,

$$\theta_r > \theta_i.$$

- At a critical angle of incidence $\theta_i^+ = \theta_c$, the light grazes the interface. The angle of refraction is

$$\theta_r^+ = \frac{\pi}{2}$$

Total internal reflection

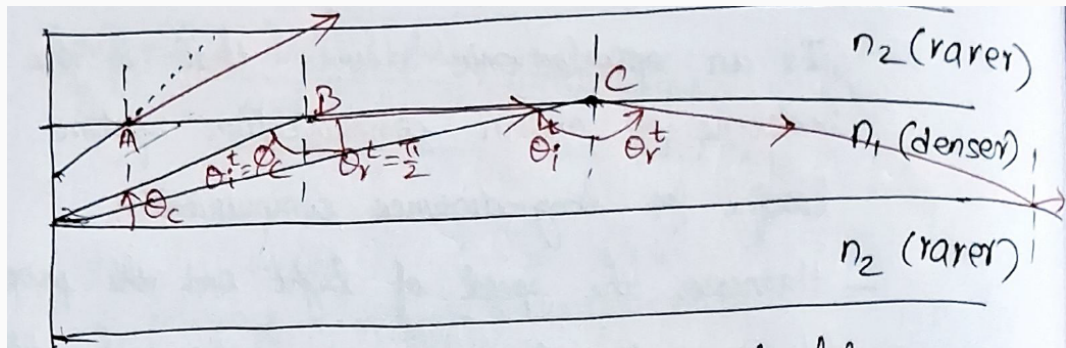


- Case (iii) For angle of incidence $\theta_i^{++} > \theta_c$, the light reflects back into the medium. The angle of reflection

$$\theta_r^{++} = \theta_i^{++}.$$

- Since the light enters the medium again with reflection and all the light energy is within the medium, this condition is called **Total Internal Reflection (TIR)**.
- TIR is required for light containment within the optical fiber. This possible if refractive index of core is greater than refractive index of cladding.

Optical fiber – conditions



- Light is at one end of the optical fiber.
- Not all light can be contained within the light pipe (optical fiber).

At the "top" interface

- If the angle of incidence is less than the critical angle of incidence, the light refracts and escapes the fiber. (case A)
- If the angle of incidence is equal to the critical angle of incidence, the light grazes the interface and is contained within the core. (case B)
- If the angle of incidence is greater than the critical angle of incidence, the light reflects back into the core.

At the "bottom" interface

- After first reflection, the light is incident at "bottom" interface.
- We again encounter the condition of denser to rarer medium.
- However, from arguments of plane geometry, it can be shown that

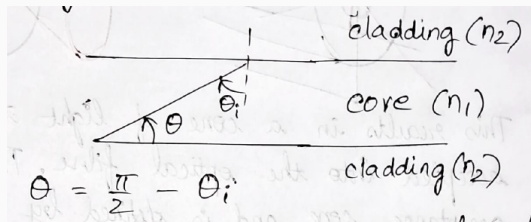
$$\theta_i^b = \theta_r^t > \phi_c$$

- So TIR occurs again. If we extend this analysis *ad infinitum*, it is proved that light is contained within the fiber for all angles of incidence greater than the critical angle of incidence.

Critical angle of propagation

Definition

Angle of propagation is defined as the angle between incident ray and core-cladding interface.



Acceptance angle

- Before the light enters the core medium, it has to encounter the interface between the core and the ambient ² medium.
- If the refractive index of ambient medium is n_3 , then for entering the core medium, the refractive index of ambient must be lesser than refractive index of core.
- For convenience, let us consider $n_3 = 1$. Therefore, the angle of incidence at the ambient-core interface needs to be less than the so called acceptance angle so that the refracted light is within the critical angle of propagation.
- At the acceptance angle of incidence θ_0 ,

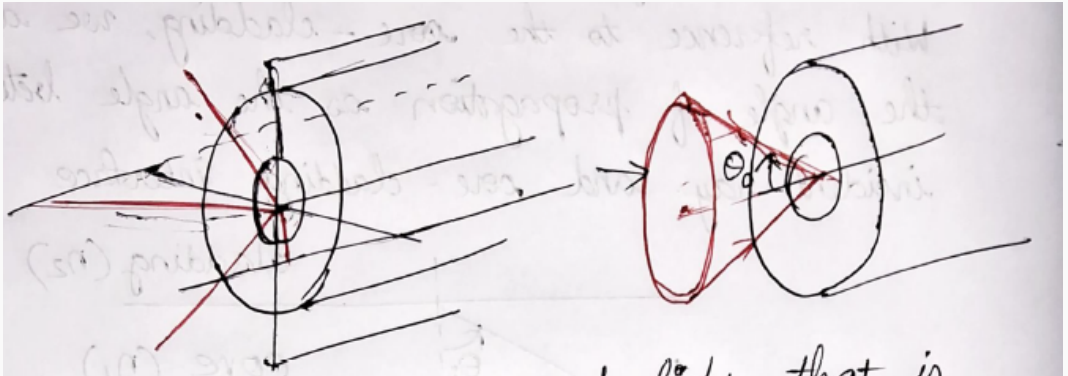
$$\frac{\sin \theta_0}{\sin \theta_0} = \frac{n_1}{n_3} = n_1$$

$$\theta_0 = \sin^{-1} \sqrt{n_1^2 - n_2^2}$$

²ambient means outside or commonly occurring, in our case it is usually air.

Acceptance cone

- We have limited our previous analysis to the "top" and "bottom" interface.
- Due to the cylindrical nature of optical fiber, the analysis is applicable to all planes passing through the axis.



This results in a cone of light that is accepted into the optical fiber. This is called acceptance cone and is defined by its acceptance angle.

- The industry has defined new quantities to quantify an optical fibre –
 1. numerical aperture
 2. fractional refractive index

Definition

Fractional refractive index is defined as relative difference in the refractive indices of core and cladding.

- The fractional refractive index is given by

$$\Delta = \frac{n_1 - n_2}{n_1}.$$

- In order to guide light effectively through a fiber, $\Delta \ll 1$. Typically, Δ is of the order of 0.01.

Numerical aperture

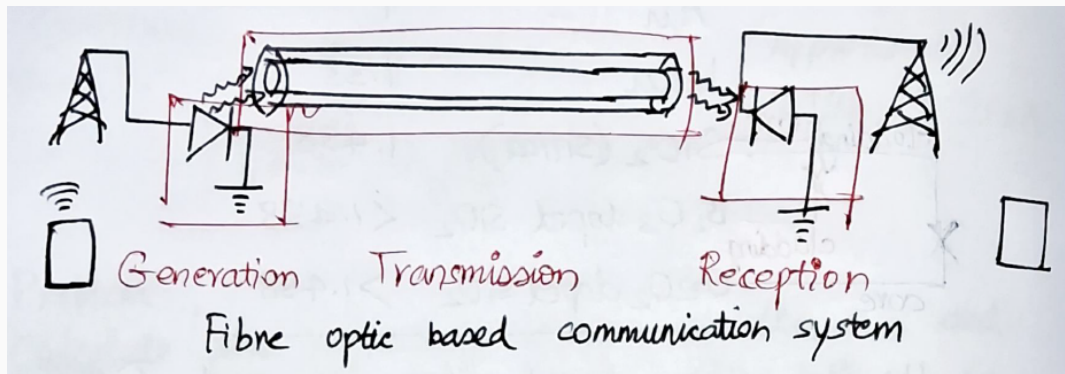
Definition

Numerical aperture is defined as the sine of the acceptance angle

$$\begin{aligned}\text{N.A.} &= \sin \theta_0, \\ &= \sin \left(\sin^{-1} \sqrt{n_1^2 - n_2^2} \right), \\ &= \sqrt{n_1^2 - n_2^2}, \\ &= \sqrt{(n_1 + n_2)(n_1 - n_2)}, \\ &= \sqrt{\left(\frac{n_1 + n_2}{2} \right) \left(\frac{n_1 - n_2}{n_1} \right)} \cdot 2n_1, \\ &\simeq \sqrt{n_1 \Delta 2n_1}, \\ &= n_1 \sqrt{2\Delta}.\end{aligned}$$

Optical fiber based communication

- Optical fiber is the backbone of modern communication system.



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